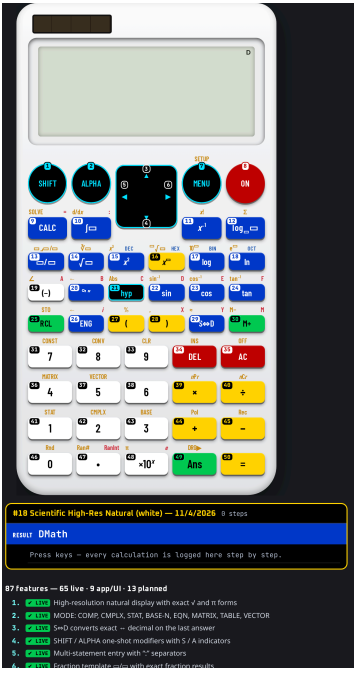
U-45 CALCULATORS

User Manual

Calculator #18 Scientific High-Res Natural (white)

Scientific – natural display



Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#18 — Scientific High-Res Natural (white)** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

How key presses are written

Keys are written as a chip showing the **key number** and its face, for example [#1 SHIFT]. Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

Entry style of this model

algebraic (type the expression as written, then = / EXE).

Chapters

1. Getting started
2. Features
3. Numbered key map
4. Key-by-key reference
5. Worked examples
6. Tricks & hacks
7. Keyboard shortcuts
8. Common mistakes
9. Practice drills
10. Appendix A — Drill answers
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12. Appendix C — Glossary

13. Appendix D — Model notes & limits

14. Appendix E — Quick-reference card

1 · Getting started

Open U-45 CALCULATORS, choose **#18** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

Tooltips

Hover over (or long-press) a key to see "#number name".

Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

Colour groups on this calculator

Colour	Group	Keys on this model
num	Digits & number entry	#19, #31, #32, #33, #36, #37, #38, #41, #42, #43, #46, #47, #48
op	Operators & execute	#16, #27, #28, #39, #40, #44, #45, #50
fn	Functions	#9, #10, #11, #12, #13, #14, #15, #17, #18, #20, #22, #23, #24, #26, #29
mem	Memory, variables & registers	#25, #30, #49
clr	Clear, delete & power	#8, #34, #35

mod	**Shift, modes, menus & navigation**	**#1, #2, #3, #4, #5, #6, #7, #21**

2 · Features

87 features: 65 live on the keys, 9 app/UI features, 13 planned (documented but not yet working — they are clearly marked and never used in examples).

1. LIVE **High-resolution natural display with exact √ and π forms**
2. LIVE **MODE: COMP, CMPLX, STAT, BASE-N, EQN, MATRIX, TABLE, VECTOR**
3. LIVE **S⇔D converts exact ↔ decimal on the last answer**
4. LIVE **SHIFT / ALPHA one-shot modifiers with S / A indicators**
5. LIVE **Multi-statement entry with ":" separators**
6. LIVE **Fraction template ▭/▭ with exact fraction results**
7. LIVE **Mixed-number entry (a b/c) and improper ↔ mixed conversion**
8. LIVE **S⇔D / F⇔D toggle between exact (fraction, √, π) and decimal answers**
9. LIVE **Square root √**
10. LIVE **Cube root $\sqrt[3]{}$**
11. LIVE **Any-index root $\sqrt[x]{y}$**
12. LIVE **Square x^2**
13. LIVE **Cube x^3**
14. LIVE **Reciprocal x^{-1} (1/x)**
15. LIVE **Power x^y / ^ with right-to-left chaining (2^3^2 = 512)**
16. LIVE **Common logarithm $\log_{10}$**
17. LIVE **Natural logarithm ln**
18. LIVE **Logarithm to any base $\log_a b$**
19. LIVE **Antilog 10^x**
20. LIVE **Exponential e^x**

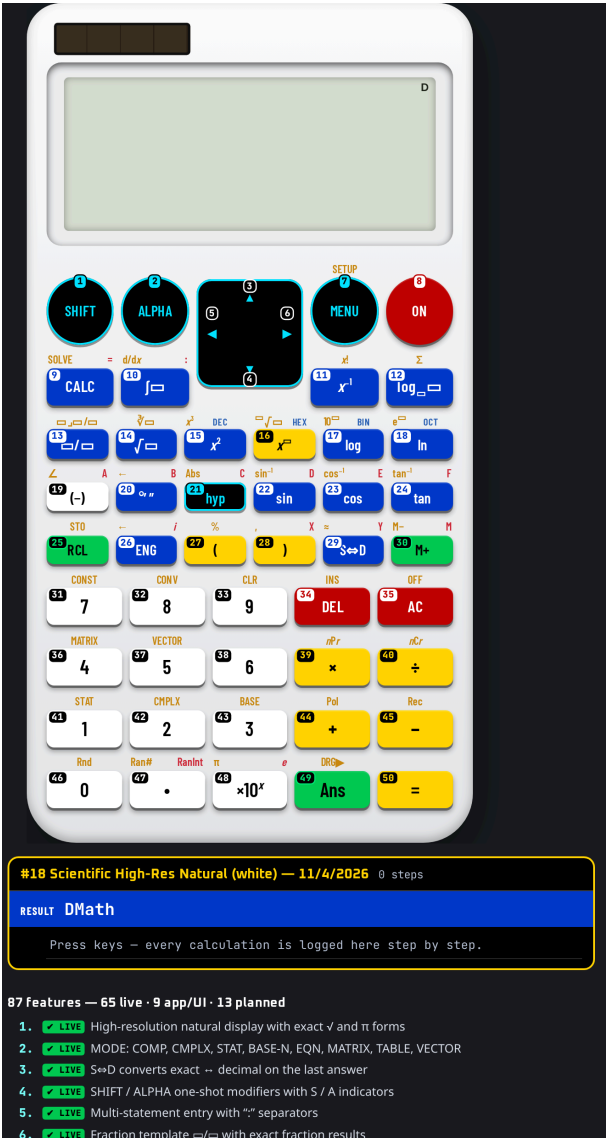
21. LIVE **Trigonometry sin / cos / tan honouring the current angle unit**
22. LIVE **Inverse trigonometry $\sin^{-1}$ / $\cos^{-1}$ / $\tan^{-1}$ with domain checking (Math ERROR outside −1…1)**
23. LIVE **Hyperbolic sinh / cosh / tanh and their inverses via hyp**
24. LIVE **Angle units DEG / RAD / GRA with unit-suffix conversion (°, r, g)**
25. LIVE **Sexagesimal ° ′ ″ entry and decimal ↔ DMS conversion**
26. LIVE **Rectangular → polar conversion Pol(x,y) → r, θ**
27. LIVE **Polar → rectangular conversion Rec(r,θ) → x, y**
28. LIVE **Factorial x! (exact up to 69!)**
29. LIVE **Permutations nPr**
30. LIVE **Combinations nCr**
31. LIVE **Random number Ran# (0 ≤ r < 1)**
32. LIVE **Random integer RanInt#(a,b)**
33. LIVE **Rnd — round the value to the displayed precision**
34. LIVE **Absolute value |x|**
35. LIVE **Percent key (x %, add-on and discount chains)**
36. LIVE **π constant to full internal precision**
37. LIVE **Euler e constant**
38. LIVE **Scientific-notation entry ×10^x / EXP / EEX**
39. LIVE **Sign change (−) / +/− / CHS**
40. LIVE **Engineering notation ENG (exponent multiple of 3)**
41. LIVE **ENG← shift the exponent the other way**
42. LIVE **Ans recall of the last result; operators auto-insert Ans**
43. LIVE **Store a value into memory / a variable (STO, sto▸, →, Min)**

44. LIVE Recall memory / variables (RCL, MR, recall)
45. LIVE Independent memory M+
46. LIVE Independent memory M−
47. LIVE Named variables (A–F, X, Y, M ... or A–Z) usable inside expressions
48. LIVE Numeric definite integral $\int_a^b f(x)dx$
49. LIVE Numeric derivative d/dx at a point
50. LIVE Summation Σ(f(x), x=a...b)
51. LIVE CALC — evaluate a stored expression for new variable values
52. LIVE SOLVE — numeric equation solver (Newton / bisection)
53. LIVE Statistics: data entry, n, $\bar{x}$, σ, s, Σx, Σx^2
54. LIVE Two-variable statistics and linear regression (a, b, r)
55. LIVE Matrix mode: define, add, multiply, det, transpose, identity
56. LIVE Vector mode: dot and cross products, magnitude
57. LIVE Complex numbers: i, arg, conjugate, Re / Im, polar ↔ rectangular
58. LIVE BASE-N: DEC / HEX / BIN / OCT conversions
59. LIVE Bitwise logic AND / OR / XOR / XNOR / NOT
60. LIVE Built-in scientific constants
61. LIVE Unit conversions (length, mass, volume, temperature)
62. LIVE MODE / SETUP: Fix / Sci / Norm display, angle unit, fraction format
63. LIVE Backspace / DEL editing of the current entry
64. LIVE INS insert mode for editing in the middle of an expression
65. LIVE Replay ▲ ▼ — scroll back to previous calculations and re-edit them
66. APP/UI Every key numbered 1...N in reading order — matches the key map in the user manual

67. APP/UI High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a “Classic” toggle
68. APP/UI Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)
69. APP/UI Show / hide key numbers with one button (remembered between sessions)
70. APP/UI Copy the current display text to the clipboard
71. APP/UI Physical-keyboard shortcuts: digits, + – * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i
72. APP/UI Fit / 1:1 zoom that never pushes keys off a phone screen
73. APP/UI Reference photo viewer and tool-tip name on every key
74. APP/UI Reset = fresh power-on state; works fully offline (PWA cache / Android APK)
75. PLANNED Spreadsheet app
76. PLANNED Verify (equation true/false)
77. PLANNED Ratio solver
78. PLANNED Inequality solver
79. PLANNED QR-code result export
80. PLANNED Prime factorisation
81. PLANNED Recurring decimals
82. PLANNED Inverse-normal distribution
83. PLANNED Exam mode (locks memory & shows indicator)
84. PLANNED Answer history export to text
85. PLANNED Speech read-out of results
86. PLANNED Colour-blind safe key theme

87. PLANNED Custom shortcut keys

3 · Numbered key map



Calculator #18 with key numbers shown (Hi-Contrast colours).

Key table

“Shift” = 2nd / f / SHIFT layer; “Alpha” = ALPHA layer; “3rd” = 3rd / g layer. A dot means the layer is empty on that key.

#	Face	Shift	Alpha	3rd	Group
1	SHIFT	·	·	·	mod
2	ALPHA	·	·	·	mod
3	▲	·	·	·	mod
4	▼	·	·	·	mod
5	◀	·	·	·	mod
6	▶	·	·	·	mod
7	MENU	SETUP	·	·	mod
8	ON	·	·	·	clr
9	CALC	SOLVE	=	·	fn
10	∫▭	d/dx	:	·	fn
11	x^{-1}	x!	·	·	fn
12	log▭▭	Σ	·	·	fn
13	▭/▭	▭␣▭/▭	·	·	fn
14	√▭	$\sqrt[3]{\square}$	·	·	fn
15	x^2	x^3	·	·	fn
16	x▭	▭√▭	·	·	op
17	log	10▭	·	·	fn
18	ln	e▭	·	·	fn
19	(–)	∠	A	·	num
20	°′ ″	←	B	·	fn
21	hyp	Abs	C	·	mod
22	sin	$\sin^{-1}$	D	·	fn
23	cos	$\cos^{-1}$	E	·	fn

#	Face	Shift	Alpha	3rd	Group
24	tan	$\tan^{-1}$	F	·	fn
25	RCL	STO	·	·	mem
26	ENG	←	i	·	fn
27	(	%	·	·	op
28	)	,	X	·	op
29	S⇔D	≈	Y	·	fn
30	M+	M−	M	·	mem
31	7	CONST	·	·	num
32	8	CONV	·	·	num
33	9	CLR	·	·	num
34	DEL	INS	·	·	clr
35	AC	OFF	·	·	clr
36	4	MATRIX	·	·	num
37	5	VECTOR	·	·	num
38	6	·	·	·	num
39	×	nPr	·	·	op
40	÷	nCr	·	·	op
41	1	STAT	·	·	num
42	2	CMPLX	·	·	num
43	3	BASE	·	·	num
44	+	Pol	·	·	op
45	−	Rec	·	·	op
46	0	Rnd	·	·	num
47	•	Ran#	RanInt	·	num
48	×10x	π	e	·	num
49	Ans	DRG▶	·	·	mem
50	=	·	·	·	op

4 · Key-by-key reference

Every key on calculator #18, in number order, with every function it carries in this app.

#1 SHIFT `Shift, modes, menus & navigation`

Primary: SHIFT / 2nd — the next key uses its upper (shift-coloured) function

#2 ALPHA `Shift, modes, menus & navigation`

Primary: ALPHA — the next key types its letter/variable (red/green label)

#3 ▲ `Shift, modes, menus & navigation`

Primary: Cursor ▲ — up / previous history line / previous item

#4 ▼ `Shift, modes, menus & navigation`

Primary: Cursor ▼ — down / next history line / next item

#5 ◀ `Shift, modes, menus & navigation`

Primary: Cursor ◀ — move left in the line

#6 ▶ `Shift, modes, menus & navigation`

Primary: Cursor ▶ — move right in the line

#7 MENU `Shift, modes, menus & navigation`

Primary: MODE — choose calculation mode

shifted (2nd / f / SHIFT layer) (press #1 first): SETUP — SETUP — angle unit, number format, display options

#8 ON `Clear, delete & power`

Primary: ON — power on

#9 CALC `Functions`

Primary: CALC — evaluate a stored formula for new variable values

shifted (2nd / f / SHIFT layer) (press #1 first): SOLVE — SOLVE — numerically solve an equation for a variable

alpha layer (press #2 first): = — Equals — complete the calculation

#10 ∫▭ `Functions`

Primary: Definite integral ∫ (numerical, Simpson/adaptive)

shifted (2nd / f / SHIFT layer) (press #1 first): d/dx — Derivative d/dx at a point (numerical)

alpha layer (press #2 first): : — Colon : (multi-statement separator)

#11 x^{-1} Functions

Primary: Reciprocal x^{-1} (1/x)

shifted (2nd / f / SHIFT layer) (press #1 first): x! — Factorial n! (also on non-integers via the gamma function where supported)

Keyboard: i

#12 log▭▭ Functions

Primary: log with any base: $\log_a(b)$

shifted (2nd / f / SHIFT layer) (press #1 first): Σ — Summation Σ over an integer range

#13 ▭/▭ Functions

Primary: Fraction template ▭/▭ (numerator over denominator)

shifted (2nd / f / SHIFT layer) (press #1 first): ▭⌟▭/▭ — Mixed-number template (whole, numerator, denominator)

#14 √▭ Functions

Primary: Square root √

shifted (2nd / f / SHIFT layer) (press #1 first): $\sqrt[3]{\ }$▭ — Cube root $\sqrt[3]{\ }$

Keyboard: r

#15 x^2 Functions

Primary: Square x^2

shifted (2nd / f / SHIFT layer) (press #1 first): x^3 — Cube x^3

Keyboard: q

#16 x▭ Operators & execute

Primary: Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)

shifted (2nd / f / SHIFT layer) (press #1 first): $\sqrt[▭]{▭}$ — n-th root $\sqrt[x]{y}$

Keyboard: ^

#17 log Functions

Primary: Common logarithm $\log_{10}$

shifted (2nd / f / SHIFT layer) (press #1 first): $10^{▭}$ — 10^x — antilog

Keyboard: l

#18 ln Functions

Primary: Natural logarithm ln

shifted (2nd / f / SHIFT layer) (press #1 first): $e^{▭}$ — e^x — exponential

Keyboard: n

#19 (−) Digits & number entry

Primary: Negative sign (−) — makes the next number negative. Different from the subtraction key

shifted (2nd / f / SHIFT layer) (press #1 first): ∠ — Angle symbol ∠ (polar complex form)

alpha layer (press #2 first): A — Variable A (store with STO/ →, recall by name)

#20 °′ ″ Functions

Primary: Degrees-minutes-seconds entry/display °′ ″

shifted (2nd / f / SHIFT layer) (press #1 first): ← — Degrees-minutes-seconds entry/display °′ ″

alpha layer (press #2 first): B — Variable B (store with STO/ →, recall by name)

#21 hyp Shift, modes, menus & navigation

Primary: Hyperbolic prefix — the next sin/cos/tan becomes sinh/cosh/tanh

shifted (2nd / f / SHIFT layer) (press #1 first): Abs — Absolute value |x|

alpha layer (press #2 first): C — Variable C (store with STO/ →, recall by name)

#22 sin Functions

Primary: Sine (uses the current angle unit)

shifted (2nd / f / SHIFT layer) (press #1 first): $\sin^{-1}$ — Inverse sine $\sin^{-1}$

alpha layer (press #2 first): D — Variable D (store with STO/ →, recall by name)

Keyboard: S

#23 cos Functions

Primary: Cosine (current angle unit)

shifted (2nd / f / SHIFT layer) (press #1 first): $\cos^{-1}$ — Inverse cosine $\cos^{-1}$

alpha layer (press #2 first): E — Variable E (store with STO/ →, recall by name)

Keyboard: c

#24 tan Functions

Primary: Tangent (current angle unit)

shifted (2nd / f / SHIFT layer) (press #1 first): $\tan^{-1}$ — Inverse tangent $\tan^{-1}$

alpha layer (press #2 first): F — Variable F (store with STO/ →, recall by name)

Keyboard: t

#25 RCL Memory, variables & registers

Primary: RCL — recall a memory / variable

shifted (2nd / f / SHIFT layer) (press #1 first): STO — STO — store the value into a memory / variable (press the target next)

#26 ENG Functions

Primary: ENG — engineering notation (exponent multiple of 3)

shifted (2nd / f / SHIFT layer) (press #1 first): ← — ←ENG — engineering notation, shifting the other way

alpha layer (press #2 first): i — Imaginary unit i (complex numbers)

#27 (Operators & execute

Primary: Opening bracket (

shifted (2nd / f / SHIFT layer) (press #1 first): % — Percent %

Keyboard: (

#28) Operators & execute

Primary: Closing bracket). Closing brackets at the end of a line may be omitted

shifted (2nd / f / SHIFT layer) (press #1 first): , — Comma / argument separator (between function arguments, list items or coordinates)

alpha layer (press #2 first): X — Variable X (store with STO/ →, recall by name)

Keyboard:)

#29 S⇔D Functions

Primary: S⇔D — toggle the result between exact (fraction, √, π) and decimal

shifted (2nd / f / SHIFT layer) (press #1 first): ≈ — ▶Decimal — show the result as a decimal

alpha layer (press #2 first): Y — Variable Y (store with STO/ →, recall by name)

#30 M+ Memory, variables & registers

Primary: M+ — add the value to independent memory M

shifted (2nd / f / SHIFT layer) (press #1 first): M− — M− — subtract the value from memory M

alpha layer (press #2 first): M — Variable M (store with STO/ →, recall by name)

#31 7 Digits & number entry

Primary: Digit 7

shifted (2nd / f / SHIFT layer) (press #1 first): CONST — CONST — scientific-constants menu

Keyboard: 7

#32 8 Digits & number entry

Primary: Digit 8

shifted (2nd / f / SHIFT layer) (press #1 first): CONV — CONV — unit-conversion menu

Keyboard: 8

#33 9 Digits & number entry

Primary: Digit 9

shifted (2nd / f / SHIFT layer) (press #1 first): CLR — CLR — clear menu (memory / setup / all)

Keyboard: 9

#34 DEL Clear, delete & power

Primary: DEL — delete the character at the cursor

shifted (2nd / f / SHIFT layer) (press #1 first): INS — INS — insert mode at the cursor

Keyboard: Backspace Delete

#35 AC `Clear, delete & power`

Primary: ON / AC — power on, or clear all

shifted (2nd / f / SHIFT layer) (press #1 first): OFF — OFF — power off (state is lost; press ON to restart)

#36 4 `Digits & number entry`

Primary: Digit 4

shifted (2nd / f / SHIFT layer) (press #1 first): MATRIX — MATRIX menu

Keyboard: 4

#37 5 `Digits & number entry`

Primary: Digit 5

shifted (2nd / f / SHIFT layer) (press #1 first): VECTOR — VECTOR menu

Keyboard: 5

#38 6 `Digits & number entry`

Primary: Digit 6

Keyboard: 6

#39 × `Operators & execute`

Primary: Multiplication ×

shifted (2nd / f / SHIFT layer) (press #1 first): nPr — Permutations nPr

Keyboard: * x

#40 ÷ `Operators & execute`

Primary: Division ÷

shifted (2nd / f / SHIFT layer) (press #1 first): nCr — Combinations nCr

Keyboard: /

#41 1 `Digits & number entry`

Primary: Digit 1

shifted (2nd / f / SHIFT layer) (press #1 first): STAT — STAT — statistics menu

Keyboard: 1

#42 2 `Digits & number entry`

Primary: Digit 2

shifted (2nd / f / SHIFT layer) (press #1 first): CMPLX — CMPLX — complex-number menu

Keyboard: 2

#43 3 `Digits & number entry`

Primary: Digit 3

shifted (2nd / f / SHIFT layer) (press #1 first): BASE — BASE-N — number-base menu (DEC/HEX/BIN/OCT)

Keyboard: 3

#44 **+** `Operators & execute`

Primary: Addition +

shifted (2nd / f / SHIFT layer) (press #1 first): Pol — Rectangular → polar Pol(x,y)

Keyboard: `+`

#45 **–** `Operators & execute`

Primary: Subtraction –

shifted (2nd / f / SHIFT layer) (press #1 first): Rec — Polar → rectangular Rec(r,θ)

Keyboard: `-`

#46 **0** `Digits & number entry`

Primary: Digit 0

shifted (2nd / f / SHIFT layer) (press #1 first): Rnd — Round to the displayed precision

Keyboard: `0`

#47 **•** `Digits & number entry`

Primary: Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored

shifted (2nd / f / SHIFT layer) (press #1 first): Ran# — Random number 0–1

alpha layer (press #2 first): RanInt — Random integer in a range

Keyboard: `.` `,`

#48 ×10x `Digits & number entry`

Primary: ×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23

shifted (2nd / f / SHIFT layer) (press #1 first): π — π (3.14159265…) as an exact constant

alpha layer (press #2 first): e — Euler's number e (2.71828…)

Keyboard: `e`

#49 Ans `Memory, variables & registers`

Primary: Ans — recall the last answer into the current expression

shifted (2nd / f / SHIFT layer) (press #1 first): DRG▶ — DRG — choose/convert the angle unit (Deg / Rad / Grad)

#50 = `Operators & execute`

Primary: Execute / = — evaluate the line and show the result (EXE / ENTER / =)

Keyboard: `Enter` `=`

5 · Worked examples

21 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.

Example 5.1 — Order of operations

Calculate 12 + 3 × 4. Press Reset first.

#41 1 → #42 2 → #44 + → #43 3 → #39 × → #36 4 → #50 =

Display shows 24 **— expected 24. ✔ verified on the app.**

Multiplication is done before addition, so the calculator holds the + until the × part is finished.

Example 5.2 — Brackets first

Calculate (12 + 8) ÷ 4. Press Reset first.

#27 (→ #41 1 → #42 2 → #44 + → #32 8 → #28) → #40 ÷ → #36 4 → #50 =

Display shows 5 **— expected 5. ✔ verified on the app.**

Brackets force the sum to be worked out before the division.

Example 5.3 — Two brackets multiplied

Calculate (3 + 4) × (9 − 2). Press Reset first.

#27 (→ #43 3 → #44 + → #36 4 → #28) → #39 × → #27 (→ #33 9 → #45 − → #42 2 → #28) → #50 =

Display shows 49 **— expected 49. ✔ verified on the app.**

Both brackets are evaluated first, then multiplied.

Example 5.4 — Squares

Calculate 12^2. Press Reset first.

#41 1 → #42 2 → #15 x^2 → #50 =

Display shows 144 **— expected 144. ✔ verified on the app.**

x^2 squares the value — quicker and safer than typing the number twice.

Example 5.5 — Square roots

Calculate √144. Press Reset first.

#14 √▭ → #41 1 → #36 4 → #36 4 → #50 =

Display shows 12 **— expected 12. ✔ verified on the app.**

The square-root key returns the principal (positive) root.

Example 5.6 — Powers

Calculate **2^10**. Press **Reset** first.

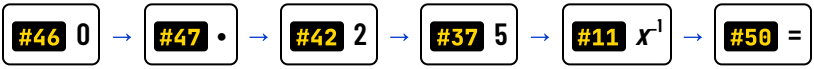


Display shows `1024` — expected 1024. ✔ verified on the app.

The power key raises the first number to the second.

Example 5.7 — Reciprocals

Calculate **1 ÷ 0.25 (x^{-1})**. Press **Reset** first.



Display shows `4` — expected 4. ✔ verified on the app.

x^{-1} (1/x) gives the reciprocal — handy for resistors in parallel or rates.

Example 5.8 — Trigonometry (degrees)

Calculate **tan 45° (Deg mode)**. Press **Reset** first.



Display shows `1` — expected 1. ✔ verified on the app.

Trig functions use the current angle unit; these answers assume degrees (the start-up setting).

Example 5.9 — Common logarithm

Calculate **log 1000**. Press **Reset** first.

#17 log → #41 1 → #46 0 → #46 0 → #46 0 → #28) → #50 =

Display shows 3 — expected 3. ✔ verified on the app.

log is base-10: it counts the zeros of a power of ten.

Example 5.10 — Natural logarithm

Calculate **ln 1**. Press **Reset** first.

#18 ln → #41 1 → #28) → #50 =

Display shows 0 — expected 0. ✔ verified on the app.

ln is the natural (base-e) logarithm.

Example 5.11 — Negative numbers

Calculate **−5 + 12**. Press **Reset** first.

#19 (−) → #37 5 → #44 + → #41 1 → #42 2 → #50 =

Display shows 7 — expected 7. ✔ verified on the app.

Use the sign key for a negative number — not the subtraction key.

Example 5.12 — Scientific notation

Calculate 6×10^2 × 3. Press Reset first.

#38 6 → #48 ×10^x → #42 2 → #39 × → #43 3 → #50 =

Display shows 1800 **— expected 1800. ✔ verified on the app.**

The exponent key enters ×10^n without typing × 10 ^.

Example 5.13 — Area of a circle πr²

Calculate π × 2.5². Press Reset first.

#1 SHIFT → #48 π → #39 × → #42 2 → #47 • → #37 5 → #15 x² → #50 =

Display shows 19.63495408 **— expected 19.63495408. ✔ verified on the app.**

π is stored to full precision — never type 3.14.

Example 5.14 — Combinations nCr

Calculate 5C2. Press Reset first.

#37 5 → #1 SHIFT → #40 nCr → #42 2 → #50 =

Display shows 10 **— expected 10. ✔ verified on the app.**

nCr counts selections where order does not matter.

Example 5.15 — Factorials

Calculate 5!. Press Reset first.

#37 5 → #1 SHIFT → #11 *x!* → #50 =

Display shows 120 **— expected 120. ✔ verified on the app.**

n! multiplies 1×2×…×n; 0! is 1 by definition.

Example 5.16 — Continue from the answer

Calculate 100 × 4 = , then Ans ÷ 2. Press Reset first.

#41 1 → #46 0 → #46 0 → #39 × → #36 4 → #50 = →
#40 ÷ → #42 2 → #50 =

Display shows 200 **— expected 200. ✔ verified on the app.**

Pressing an operator straight after a result continues from that answer (Ans).

Example 5.17 — Percent of an amount

Calculate 15 % of 80. Press Reset first.

#32 8 → #46 0 → #39 × → #41 1 → #37 5 → #1 SHIFT →
#27 % → #50 =

Display shows 12 **— expected 12. ✔ verified on the app.**

The percent key turns the second number into a percentage of the first.

Example 5.18 — Average of three numbers

Calculate **(4 + 8 + 15) ÷ 3**. Press **Reset** first.

#27 (→ **#36** 4 → **#44** + → **#32** 8 → **#44** + → **#41** 1 → **#37** 5 → **#28**) → **#40** ÷ → **#43** 3 → **#50** =

Display shows `9` — expected 9. ✔ verified on the app.

Add up the values, then divide by how many there are.

Example 5.19 — Mixed operations

Calculate **50 − 4 × 6 + 7**. Press **Reset** first.

#37 5 → **#46** 0 → **#45** − → **#36** 4 → **#39** × → **#38** 6 → **#44** + → **#31** 7 → **#50** =

Display shows `33` — expected 33. ✔ verified on the app.

Only the × part is done first; the − and + are then worked left to right.

Example 5.20 — Dividing twice

Calculate **360 ÷ 4 ÷ 9**. Press **Reset** first.

#43 3 → **#38** 6 → **#46** 0 → **#40** ÷ → **#36** 4 → **#40** ÷ → **#33** 9 → **#50** =

Display shows `10` — expected 10. ✔ verified on the app.

Division is worked left to right: the first quotient is divided again.

Example 5.21 — Brackets inside a longer formula

Calculate ((3 + 5) × 4 − 2) ÷ 10. Press Reset first.

#27 (→ #27 (→ #43 3 → #44 + → #37 5 → #28) → #39 × → #36 4 → #45 − → #42 2 → #28) → #40 ÷ → #41 1 → #46 0 → #50 =

Display shows 3 **— expected 3. ✔ verified on the app.**

The innermost brackets are worked out first, then the outer brackets, then the division.

6 · Tricks & hacks

Hack 1: Edit instead of retyping

After a result, press ◀ or ▶ to bring the last expression back with the cursor in it; change one number and press = again.

Hack 2: Ans chains

Start a line with an operator (e.g. ×) and the calculator inserts Ans automatically — great for compound growth: 1.05 ×, then = = = …

Hack 3: Exact vs decimal

Natural-display models keep fractions, √ and π exact. Press S⇔D to flip to decimal; press it again to return.

Hack 4: Close brackets for free

Trailing closing brackets can be left off; = closes them for you.

Hack 5: Implicit multiplication

Type 2π or 3(4+1) without ×. Implicit multiplication binds tighter than ÷ on textbook models — use brackets if in doubt.

Hack 6: Store constants

Store a value you reuse (e.g. g = 9.81) with STO and a letter; then type the letter in formulas.

Hack 7: Power chains

x^y chains right to left: 2^3^2 = 2^9 = 512, not 64. Use brackets: (2^3)^2 = 64.

Hack 8: Angles

Check the D / R / G indicator before trig. sin 30 = 0.5 only in degrees.

Hack 9: Memory sums

M+ / M− keep a running total separate from the calculation line; RCL M shows it.

Hack 10: Replay history

Use ▲ ▼ to scroll earlier lines; press = on an old line to re-run it.

Hack 11: Hidden functions on this model

55 extra functions sit on shifted layers. Press the layer key once, then the key: for example #1 then #7 → SETUP; #1 then #9 → SOLVE; #2 then #9 → =; #1 then #10 → d/dx.

Hack 12: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left → right, top → bottom from #1 to #50.

Hack 13: Type faster on a computer

29 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

Hack 14: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

Hack 15: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

Hack 16: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

7 · Keyboard shortcuts

With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.

Keyboard	#	Key	Does
i	#11	x^{-1}	Reciprocal x^{-1} (1/x)
r	#14	$\sqrt{\square}$	Square root $\sqrt{}$
q	#15	x^2	Square x^2
^	#16	$x\square$	Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)
l	#17	log	Common logarithm $\log_{10}$
n	#18	ln	Natural logarithm ln
s	#22	sin	Sine (uses the current angle unit)
c	#23	cos	Cosine (current angle unit)
t	#24	tan	Tangent (current angle unit)
(	#27	(	Opening bracket (
)	#28	)	Closing bracket). Closing brackets at the end of a line may be omitted
7	#31	7	Digit 7
8	#32	8	Digit 8
9	#33	9	Digit 9
Backspace Delete	#34	DEL	DEL — delete the character at the cursor
4	#36	4	Digit 4
5	#37	5	Digit 5
6	#38	6	Digit 6
* x	#39	×	Multiplication ×
/	#40	÷	Division ÷

1	#41	1	Digit 1
2	#42	2	Digit 2
3	#43	3	Digit 3
+	#44	+	Addition +
-	#45	−	Subtraction −
0	#46	0	Digit 0
. ,	#47	•	Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored
e	#48	×10x	×10^{x} / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10^{x} 23
Enter =	#50	=	Execute / = — evaluate the line and show the result (EXE / ENTER / =)

App shortcuts

“# On / # Off”	show / hide key numbers
“Classic / Hi-Contrast”	switch key colours
“Copy”	copy display text
“Fit / 1:1”	scale to screen / design size
“Reset”	restart this calculator
Jump list	switch to another of the 45 calculators

8 · Common mistakes

Mistake 1: Using – instead of (–)

The negative key (–) and the subtraction key are different; "– 5" at the start uses Ans – 5.

Mistake 2: Wrong angle unit

If sin 30 gives –0.988, the calculator is in radians. Switch to degrees in SETUP.

Mistake 3: -3^2 vs $(-3)^2$

$-3^2 = -9$ because the square is applied first. Type $(-3)^2$ to get 9.

Mistake 4: Division precedence

$1 \div 2\pi$ may be read as $1 \div (2\pi)$ on textbook models. Type $1 \div (2 \times \pi)$ explicitly to be sure.

Mistake 5: Forgetting SHIFT is one-shot

SHIFT applies only to the next key. If the S indicator is still lit, press SHIFT again to cancel.

Mistake 6: Clearing memory by accident

AC clears the line but not memory; CLR (SHIFT 9) can clear memory and variables.

Mistake 7: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

Mistake 8: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

Mistake 9: Minus vs negative

The (−) negative key and the − subtraction key are different. Use (−) to start a negative number, − between two numbers.

9 · Practice drills

Set A — Calculation drills (109)

Work each one on calculator #18, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.

1. 25 + 5 × 6 = _____
2. 7 + 8 × 9 = _____
3. 150 + 12 × 11 = _____
4. 3 + 14 × 15 = _____
5. 48 + 6 × 7 = _____
6. 1000 + 25 × 4 = _____
7. 19 + 21 × 3 = _____
8. 64 + 8 × 8 = _____
9. (45 + 15) ÷ 6 = _____
10. (9 + 7) ÷ 2 = _____
11. (100 + 60) ÷ 8 = _____
12. (33 + 17) ÷ 5 = _____
13. (250 + 350) ÷ 12 = _____
14. (5 + 6) × (10 − 3) = _____
15. (12 + 3) × (8 − 5) = _____
16. (1.5 + 2.5) × (7 − 4) = _____

17. (20 + 30) × (15 − 5) = _____

18. (9 + 1) × (100 − 1) = _____

19. (11 + 13) × (17 − 9) = _____

20. 25^2 = _____

21. 7^2 = _____

22. 1.5^2 = _____

23. 19^2 = _____

24. 33^2 = _____

25. 0.5^2 = _____

26. 125^2 = _____

27. 64^2 = _____

28. $\sqrt{625}$ = _____

29. $\sqrt{1024}$ = _____

30. $\sqrt{9801}$ = _____

31. 3^5 = _____

32. 1.05^3 = _____

33. 5^4 = _____

34. 7^3 = _____

35. 1.5^2 = _____

36. 10^6 = _____

37. 2^0.5 = _____

38. 1 ÷ 0.2 (x^{-1}) = _____

39. sin 90° (Deg mode) = _____

40. cos 0° (Deg mode) = _____

41. log 100 = _____

42. log 10000 = _____

43. ln 10 = _____

44. ln 2 = _____

45. −20 + 7 = _____

46. −3 + 3 = _____

47. −100 + 250 = _____

48. −8 + 1 = _____

49. 2×10^3 × 4 = _____

50. 1.5×10^4 × 2 = _____

51. 3×10^5 × 3 = _____

52. 4.5×10^2 × 2 = _____

53. π × 1.2^2 = _____

54. 10C3 = _____

55. 6C6 = _____

56. 6! = _____

57. 0! = _____

58. 15 × 6 = , then Ans ÷ 9 = _____

59. 7.5 × 8 = , then Ans ÷ 3 = _____

60. 250 × 12 = , then Ans ÷ 100 = _____

61. 8 % of 250 = _____

62. 25 % of 360 = _____

63. 12.5 % of 40 = _____

64. 10 % of 450 = _____

65. 20 % of 125 = _____

66. 5 % of 640 = _____

67. 75 % of 48 = _____

68. 30 % of 90 = _____

69. (12 + 15 + 18) ÷ 3 = _____

70. (70 + 85 + 91) ÷ 3 = _____

71. (2.5 + 3.5 + 6) ÷ 3 = _____

72. (100 + 200 + 600) ÷ 3 = _____

73. 100 − 3 × 12 + 9 = _____

74. 18 − 2 × 5 + 40 = _____

75. 7.5 − 1.5 × 3 + 2 = _____

76. 64 − 8 × 4 + 10 = _____

77. 300 − 25 × 4 + 55 = _____

78. 9 − 3 × 3 + 1 = _____

79. 45 − 5 × 5 + 15 = _____

80. 1000 ÷ 8 ÷ 5 = _____

81. 96 ÷ 2 ÷ 3 = _____

82. 7200 ÷ 60 ÷ 60 = _____

83. 500 ÷ 4 ÷ 25 = _____

84. 81 ÷ 3 ÷ 3 = _____

85. ((10 + 2) × 3 − 6) ÷ 6 = _____

86. ((1.5 + 2.5) × 6 − 4) ÷ 5 = _____

87. ((20 + 30) × 2 − 25) ÷ 3 = _____

88. ((7 + 8) × 9 − 15) ÷ 12 = _____

89. 12 + 3 × 4 = _____

90. (12 + 8) ÷ 4 = _____

91. (3 + 4) × (9 − 2) = _____

92. 12^2 = _____

93. $\sqrt{144}$ = _____

94. 2^10 = _____

95. 1 ÷ 0.25 (x^{-1}) = _____

96. tan 45° (Deg mode) = _____

97. log 1000 = _____

98. ln 1 = _____

99. −5 + 12 = _____

100. 6×10^2 × 3 = _____

101. π × 2.5^2 = _____

102. 5C2 = _____

103. 5! = _____

104. 100 × 4 = , then Ans ÷ 2 = _____

105. 15 % of 80 = _____

106. (4 + 8 + 15) ÷ 3 = _____

107. 50 − 4 × 6 + 7 = _____

108. 360 ÷ 4 ÷ 9 = _____

109. ((3 + 5) × 4 − 2) ÷ 10 = _____

Set B — Key-finding drills (95)

Without looking at the table, find each key on the picture. Answers in Appendix A.

1. Which key is “SHIFT” (SHIFT / 2nd)? _____
2. Which key is “ALPHA” (ALPHA)? _____
3. Which key is “▲” (Cursor ▲)? _____
4. Which key is “▼” (Cursor ▼)? _____
5. Which key is “◀” (Cursor ◀)? _____

6. Which key is “▶” (Cursor ▶)? _____

7. Which key is “MENU” (MODE)? _____

8. Which key is “ON” (ON)? _____

9. Which key is “CALC” (CALC)? _____

10. Which key is “∫▭” (Definite integral ∫)? _____

11. Which key is “x^{-1}” (Reciprocal x^{-1})? _____

12. Which key is “log▭▭” (log with any base: $\log_a$)? _____

13. Which key is “▭/▭” (Fraction template ▭/▭)? _____

14. Which key is “√▭” (Square root √)? _____

15. Which key is “x^2” (Square x^2)? _____

16. Which key is “x▭” (Power x^y / ^)? _____

17. Which key is “log” (Common logarithm $\log_{10}$)? _____

18. Which key is “ln” (Natural logarithm ln)? _____

19. Which key is “(−)” (Negative sign)? _____

20. Which key is “°′ ″” (Degrees-minutes-seconds entry/display °′ ″)? _____

21. Which key is “hyp” (Hyperbolic prefix)? _____

22. Which key is “sin” (Sine)? _____

23. Which key is “cos” (Cosine)? _____

24. Which key is “tan” (Tangent)? _____

25. Which key is “RCL” (RCL)? _____

26. Which key is “ENG” (ENG)? _____

27. Which key is “(” (Opening bracket)? _____

28. Which key is “)” (Closing bracket). Closing brackets at the end of a line may be omitted)? _____

29. Which key is “S⇔D” (S⇔D)? _____

30. Which key is “M+” (M+)? _____

31. Which key is “DEL” (DEL)? _____

32. Which key is “AC” (ON / AC)? _____

33. Which key is “×” (Multiplication ×)? _____

34. Which key is “÷” (Division ÷)? _____

35. Which key is “+” (Addition +)? _____

36. Which key is “−” (Subtraction −)? _____

37. Which key is “•” (Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored)? _____

38. Which key is “×10x” (×10^{x} / EXP / EE)? _____

39. Which key is “Ans” (Ans)? _____

40. Which key is “=” (Execute / =)? _____

41. How do you reach “SETUP”? _____

42. How do you reach “SOLVE”? _____

43. How do you reach “=”? _____

44. How do you reach “d/dx”? _____

45. How do you reach ":"? _____

46. How do you reach "x!"? _____

47. How do you reach "Σ"? _____

48. How do you reach "▭⌟▭/▭"? _____

49. How do you reach "$\sqrt[3]{▭}$"? _____

50. How do you reach "x^{3}"? _____

51. How do you reach "▭√▭"? _____

52. How do you reach "10▭"? _____

53. How do you reach "e▭"? _____

54. How do you reach "∠"? _____

55. How do you reach "A"? _____

56. How do you reach "←"? _____

57. How do you reach "B"? _____

58. How do you reach "Abs"? _____

59. How do you reach "C"? _____

60. How do you reach "$\sin^{-1}$"? _____

61. How do you reach "D"? _____

62. How do you reach "$\cos^{-1}$"? _____

63. How do you reach "E"? _____

64. How do you reach "$\tan^{-1}$"? _____

65. How do you reach “F”? _____

66. How do you reach “STO”? _____

67. How do you reach “ ←”? _____

68. How do you reach “i”? _____

69. How do you reach “%”? _____

70. How do you reach “,”? _____

71. How do you reach “X”? _____

72. How do you reach “≈”? _____

73. How do you reach “Y”? _____

74. How do you reach “M–”? _____

75. How do you reach “M”? _____

76. How do you reach “CONST”? _____

77. How do you reach “CONV”? _____

78. How do you reach “CLR”? _____

79. How do you reach “INS”? _____

80. How do you reach “OFF”? _____

81. How do you reach “MATRIX”? _____

82. How do you reach “VECTOR”? _____

83. How do you reach “nPr”? _____

84. How do you reach “nCr”? _____

85. How do you reach “STAT”? _____

86. How do you reach “CMPLX”? _____

87. How do you reach “BASE”? _____

88. How do you reach “Pol”? _____

89. How do you reach “Rec”? _____

90. How do you reach “Rnd”? _____

91. How do you reach “Ran#”? _____

92. How do you reach “RanInt”? _____

93. How do you reach “π”? _____

94. How do you reach “e”? _____

95. How do you reach “DRG▶”? _____

Appendix A — Drill answers

Set A

1. 55 (display `55`)

`#42` 2 → `#37` 5 → `#44` + → `#37` 5 → `#39` × → `#38` 6 → `#50` =

2. 79 (display `79`)

`#31` 7 → `#44` + → `#32` 8 → `#39` × → `#33` 9 → `#50` =

3. 282 (display `282`)

`#41` 1 → `#37` 5 → `#46` 0 → `#44` + → `#41` 1 → `#42` 2 → `#39` × → `#41` 1 → `#41` 1 → `#50` =

4. 213 (display `213`)

`#43` 3 → `#44` + → `#41` 1 → `#36` 4 → `#39` × → `#41` 1 → `#37` 5 → `#50` =

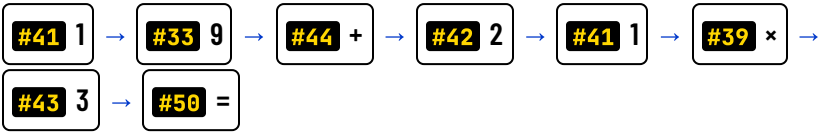
5. 90 (display `90`)

`#36` 4 → `#32` 8 → `#44` + → `#38` 6 → `#39` × → `#31` 7 → `#50` =

6. 1100 (display `1100`)

`#41` 1 → `#46` 0 → `#46` 0 → `#46` 0 → `#44` + → `#42` 2 → `#37` 5 → `#39` × → `#36` 4 → `#50` =

7. 82 (display 82)



8. 128 (display 128)



9. 10 (display 10)



10. 8 (display 8)



11. 20 (display 20)



12. 10 (display 10)



13. 50 (display 50)

#27 (→ #42 2 → #37 5 → #46 0 → #44 + → #43 3 → #37 5 → #46 0 → #28) → #40 ÷ → #41 1 → #42 2 → #50 =

14. 77 (display 77)

#27 (→ #37 5 → #44 + → #38 6 → #28) → #39 × → #27 (→ #41 1 → #46 0 → #45 − → #43 3 → #28) → #50 =

15. 45 (display 45)

#27 (→ #41 1 → #42 2 → #44 + → #43 3 → #28) → #39 × → #27 (→ #32 8 → #45 − → #37 5 → #28) → #50 =

16. 12 (display 12)

#27 (→ #41 1 → #47 • → #37 5 → #44 + → #42 2 → #47 • → #37 5 → #28) → #39 × → #27 (→ #31 7 → #45 − → #36 4 → #28) → #50 =

17. 500 (display 500)

#27 (→ #42 2 → #46 0 → #44 + → #43 3 → #46 0 → #28) → #39 × → #27 (→ #41 1 → #37 5 → #45 − → #37 5 → #28) → #50 =

18. 990 (display `990`)

`#27` (→ `#33` 9 → `#44` + → `#41` 1 → `#28`) → `#39` × →
`#27` (→ `#41` 1 → `#46` 0 → `#46` 0 → `#45` − → `#41` 1 →
`#28`) → `#50` =

19. 192 (display `192`)

`#27` (→ `#41` 1 → `#41` 1 → `#44` + → `#41` 1 → `#43` 3 →
`#28`) → `#39` × → `#27` (→ `#41` 1 → `#31` 7 → `#45` − →
`#33` 9 → `#28`) → `#50` =

20. 625 (display `625`)

`#42` 2 → `#37` 5 → `#15` x^2 → `#50` =

21. 49 (display `49`)

`#31` 7 → `#15` x^2 → `#50` =

22. 2.25 (display `2.25`)

`#41` 1 → `#47` • → `#37` 5 → `#15` x^2 → `#50` =

23. 361 (display `361`)

`#41` 1 → `#33` 9 → `#15` x^2 → `#50` =

24. 1089 (display `1089`)

`#43` 3 → `#43` 3 → `#15` x^2 → `#50` =

25. 0.25 (display `0.25`)

`#46` 0 → `#47` • → `#37` 5 → `#15` x^2 → `#50` =

26. 15625 (display `15625`)

`#41` 1 → `#42` 2 → `#37` 5 → `#15` x^2 → `#50` =

27. 4096 (display `4096`)

`#38` 6 → `#36` 4 → `#15` x^2 → `#50` =

28. 25 (display `25`)

`#14` $\sqrt{\square}$ → `#38` 6 → `#42` 2 → `#37` 5 → `#50` =

29. 32 (display `32`)

`#14` $\sqrt{\square}$ → `#41` 1 → `#46` 0 → `#42` 2 → `#36` 4 → `#50` =

30. 99 (display `99`)

`#14` $\sqrt{\square}$ → `#33` 9 → `#32` 8 → `#46` 0 → `#41` 1 → `#50` =

31. 243 (display `243`)

`#43` 3 → `#16` $x^{\square}$ → `#37` 5 → `#50` =

32. 1.157625 (display `1.157625`)

`#41` 1 → `#47` • → `#46` 0 → `#37` 5 → `#16` $x^{\square}$ → `#43` 3 → `#50` =

33. 625 (display `625`)

`#37` 5 → `#16` $x^{\square}$ → `#36` 4 → `#50` =

34. 343 (display `343`)

`#31` 7 → `#16` $x^{\square}$ → `#43` 3 → `#50` =

35. 2.25 (display `2.25`)

`#41` 1 → `#47` • → `#37` 5 → `#16` $x^{\square}$ → `#42` 2 → `#50` =

36. 1000000.000 (display `1000000`)

`#41` 1 → `#46` 0 → `#16` $x^{\square}$ → `#38` 6 → `#50` =

37. 1.41421356 (display `1.414213562`)

`#42` 2 → `#16` $x^{\square}$ → `#46` 0 → `#47` • → `#37` 5 → `#50` =

38. 5 (display `5`)

`#46` 0 → `#47` • → `#42` 2 → `#11` x^{-1} → `#50` =

39. 1 (display `1`)

`#22` sin → `#33` 9 → `#46` 0 → `#28`) → `#50` =

40. 1 (display `1`)

`#23` cos → `#46` 0 → `#28`) → `#50` =

41. 2 (display `2`)

`#17` log → `#41` 1 → `#46` 0 → `#46` 0 → `#28`) → `#50` =

42. 4 (display `4`)

`#17` log → `#41` 1 → `#46` 0 → `#46` 0 → `#46` 0 → `#46` 0 → `#28`) → `#50` =

43. 2.30258509 (display `2.302585093`)

`#18` ln → `#41` 1 → `#46` 0 → `#28`) → `#50` =

44. 0.69314718 (display `0.6931471806`)

`#18` ln → `#42` 2 → `#28`) → `#50` =

45. -13 (display `-13`)

`#19` (−) → `#42` 2 → `#46` 0 → `#44` + → `#31` 7 → `#50` =

46. 0 (display `0`)

`#19` (−) → `#43` 3 → `#44` + → `#43` 3 → `#50` =

47. 150 (display `150`)

`#19` (−) → `#41` 1 → `#46` 0 → `#46` 0 → `#44` + → `#42` 2 → `#37` 5 → `#46` 0 → `#50` =

48. -7 (display `-7`)

`#19` (−) → `#32` 8 → `#44` + → `#41` 1 → `#50` =

49. 8000 (display `8000`)

`#42` 2 → `#48` ×10^x → `#43` 3 → `#39` × → `#36` 4 → `#50` =

50. 30000 (display `30000`)

`#41` 1 → `#47` • → `#37` 5 → `#48` ×10^x → `#36` 4 → `#39` × → `#42` 2 → `#50` =

51. 900000 (display `900000`)

`#43` 3 → `#48` ×10^x → `#37` 5 → `#39` × → `#43` 3 → `#50` =

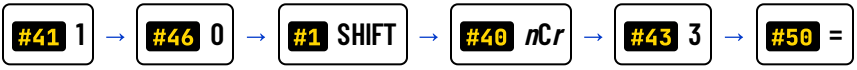
52. 900 (display `900`)



53. 4.52389342 (display `4.523893421`)



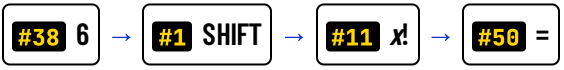
54. 120 (display `120`)



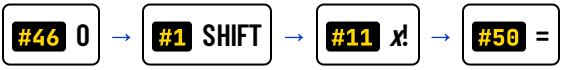
55. 1 (display `1`)



56. 720 (display `720`)



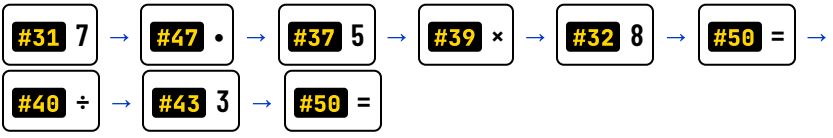
57. 1 (display `1`)



58. 10 (display `10`)



59. 20 (display `20`)



60. 30 (display 30)

#42 2 → #37 5 → #46 0 → #39 × → #41 1 → #42 2 → #50 = → #40 ÷ → #41 1 → #46 0 → #46 0 → #50 =

61. 20 (display 20)

#42 2 → #37 5 → #46 0 → #39 × → #32 8 → #1 SHIFT → #27 % → #50 =

62. 90 (display 90)

#43 3 → #38 6 → #46 0 → #39 × → #42 2 → #37 5 → #1 SHIFT → #27 % → #50 =

63. 5 (display 5)

#36 4 → #46 0 → #39 × → #41 1 → #42 2 → #47 • → #37 5 → #1 SHIFT → #27 % → #50 =

64. 45 (display 45)

#36 4 → #37 5 → #46 0 → #39 × → #41 1 → #46 0 → #1 SHIFT → #27 % → #50 =

65. 25 (display 25)

#41 1 → #42 2 → #37 5 → #39 × → #42 2 → #46 0 → #1 SHIFT → #27 % → #50 =

66. 32 (display 32)

#38 6 → #36 4 → #46 0 → #39 × → #37 5 → #1 SHIFT → #27 % → #50 =

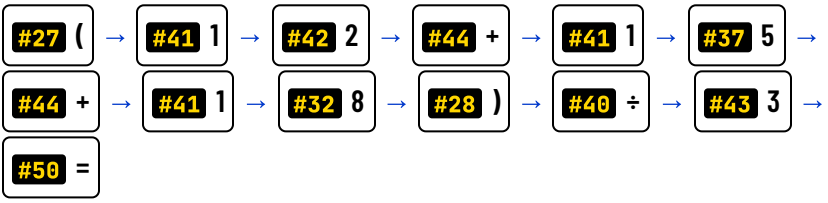
67. 36 (display `36`)



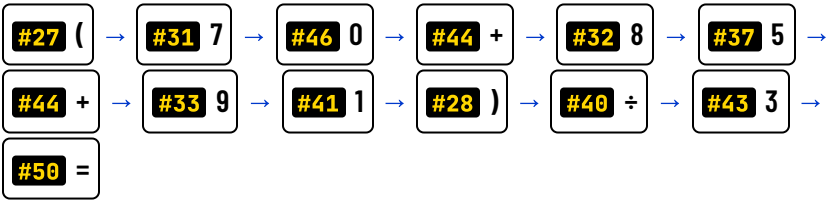
68. 27 (display `27`)



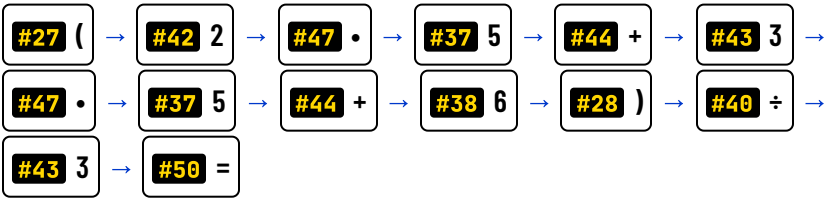
69. 15 (display `15`)



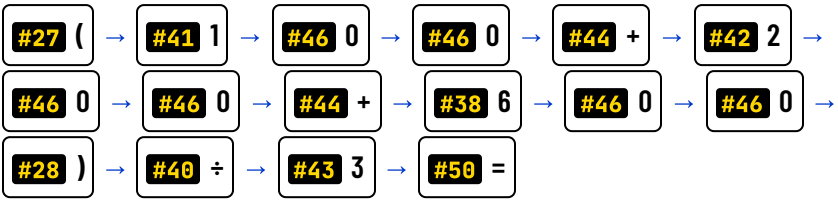
70. 82 (display `82`)



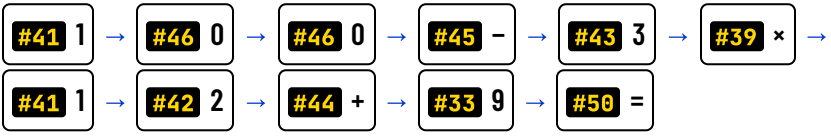
71. 4 (display `4`)



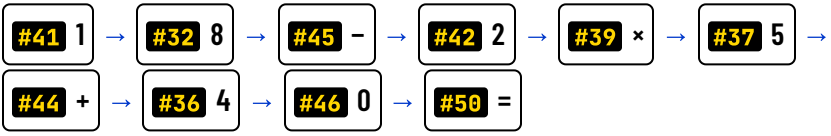
72. 300 (display 300)



73. 73 (display 73)



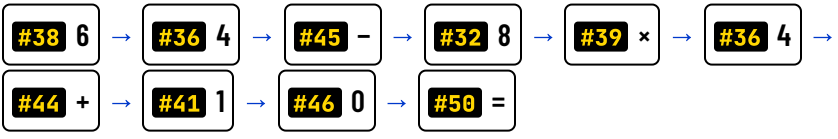
74. 48 (display 48)



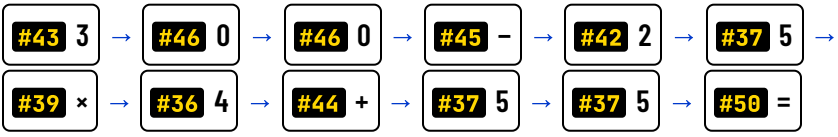
75. 5 (display 5)



76. 42 (display 42)



77. 255 (display 255)



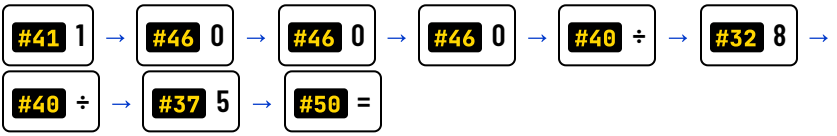
78. 1 (display `1`)



79. 35 (display `35`)



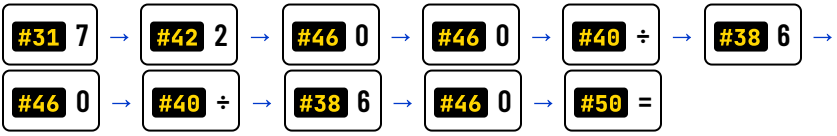
80. 25 (display `25`)



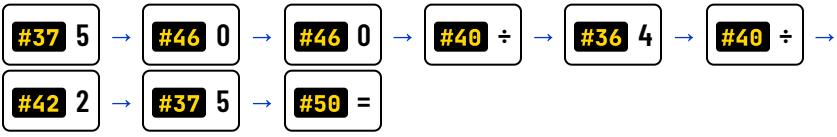
81. 16 (display `16`)



82. 2 (display `2`)



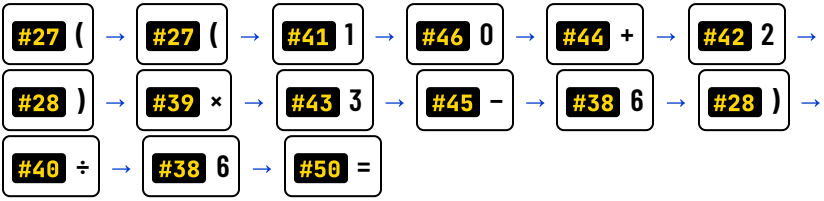
83. 5 (display `5`)



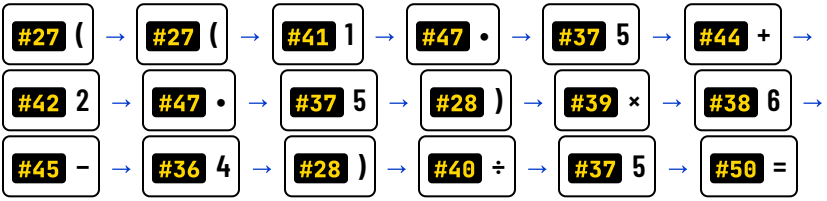
84. 9 (display `9`)



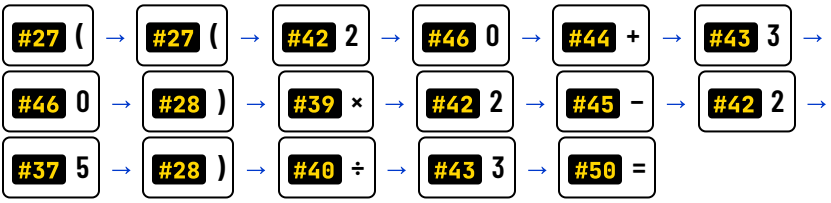
85. 5 (display 5)



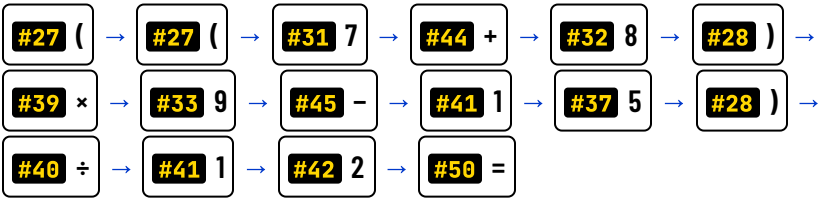
86. 4 (display 4)



87. 25 (display 25)



88. 10 (display 10)



89. 24 (display 24)



90. 5 (display `5`)

`#27` (→ `#41` 1 → `#42` 2 → `#44` + → `#32` 8 → `#28`) → `#40` ÷ → `#36` 4 → `#50` =

91. 49 (display `49`)

`#27` (→ `#43` 3 → `#44` + → `#36` 4 → `#28`) → `#39` × → `#27` (→ `#33` 9 → `#45` − → `#42` 2 → `#28`) → `#50` =

92. 144 (display `144`)

`#41` 1 → `#42` 2 → `#15` x^2 → `#50` =

93. 12 (display `12`)

`#14` $\sqrt{\square}$ → `#41` 1 → `#36` 4 → `#36` 4 → `#50` =

94. 1024 (display `1024`)

`#42` 2 → `#16` $x^{\square}$ → `#41` 1 → `#46` 0 → `#50` =

95. 4 (display `4`)

`#46` 0 → `#47` • → `#42` 2 → `#37` 5 → `#11` x^{-1} → `#50` =

96. 1 (display `1`)

`#24` tan → `#36` 4 → `#37` 5 → `#28`) → `#50` =

97. 3 (display `3`)

`#17` log → `#41` 1 → `#46` 0 → `#46` 0 → `#46` 0 → `#28`) → `#50` =

98. 0 (display `0`)

`#18` ln → `#41` 1 → `#28`) → `#50` =

99. 7 (display `7`)

`#19` (−) → `#37` 5 → `#44` + → `#41` 1 → `#42` 2 → `#50` =

100. 1800 (display `1800`)

`#38` 6 → `#48` ×10^{x} → `#42` 2 → `#39` × → `#43` 3 → `#50` =

101. 19.63495408 (display `19.63495408`)

`#1` SHIFT → `#48` π → `#39` × → `#42` 2 → `#47` • → `#37` 5 → `#15` x^2 → `#50` =

102. 10 (display `10`)

`#37` 5 → `#1` SHIFT → `#40` *n*C*r* → `#42` 2 → `#50` =

103. 120 (display `120`)

`#37` 5 → `#1` SHIFT → `#11` *x*! → `#50` =

104. 200 (display `200`)

`#41` 1 → `#46` 0 → `#46` 0 → `#39` × → `#36` 4 → `#50` = → `#40` ÷ → `#42` 2 → `#50` =

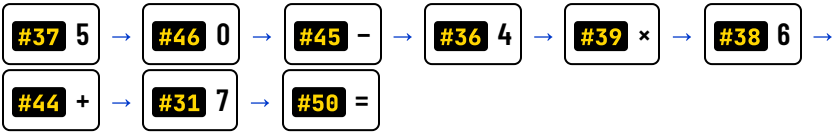
105. 12 (display `12`)

`#32` 8 → `#46` 0 → `#39` × → `#41` 1 → `#37` 5 → `#1` SHIFT → `#27` % → `#50` =

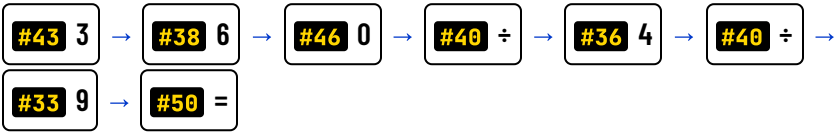
106. 9 (display `9`)

`#27` (→ `#36` 4 → `#44` + → `#32` 8 → `#44` + → `#41` 1 → `#37` 5 → `#28`) → `#40` ÷ → `#43` 3 → `#50` =

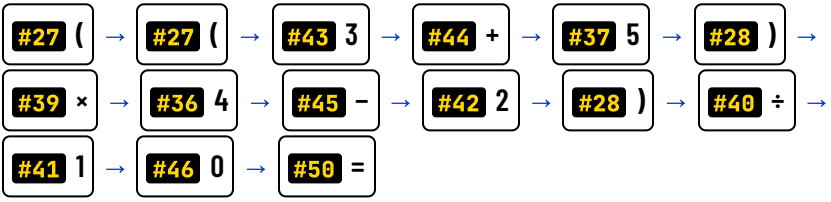
107. 33 (display `33`)



108. 10 (display `10`)



109. 3 (display `3`)



Set B

1. #1
2. #2
3. #3
4. #4
5. #5
6. #6
7. #7
8. #8
9. #9
10. #10
11. #11
12. #12
13. #13
14. #14
15. #15
16. #16
17. #17
18. #18
19. #19
20. #20
21. #21
22. #22
23. #23
24. #24
25. #25
26. #26
27. #27
28. #28
29. #29
30. #30
31. #34
32. #35
33. #39
34. #40
35. #44
36. #45
37. #47
38. #48

39. #49
40. #50
41. #1 SHIFT then #7 MENU
42. #1 SHIFT then #9 CALC
43. #2 ALPHA then #9 CALC
44. #1 SHIFT then #10 ∫▭
45. #2 ALPHA then #10 ∫▭
46. #1 SHIFT then #11 x^{-1}
47. #1 SHIFT then #12 log▭▭
48. #1 SHIFT then #13 ▭/▭
49. #1 SHIFT then #14 √▭
50. #1 SHIFT then #15 x^2
51. #1 SHIFT then #16 x▭
52. #1 SHIFT then #17 log
53. #1 SHIFT then #18 ln
54. #1 SHIFT then #19 (-)
55. #2 ALPHA then #19 (-)
56. #1 SHIFT then #20 °′″
57. #2 ALPHA then #20 °′″
58. #1 SHIFT then #21 hyp
59. #2 ALPHA then #21 hyp
60. #1 SHIFT then #22 sin
61. #2 ALPHA then #22 sin
62. #1 SHIFT then #23 cos
63. #2 ALPHA then #23 cos
64. #1 SHIFT then #24 tan
65. #2 ALPHA then #24 tan
66. #1 SHIFT then #25 RCL
67. #1 SHIFT then #26 ENG
68. #2 ALPHA then #26 ENG
69. #1 SHIFT then #27 (
70. #1 SHIFT then #28)
71. #2 ALPHA then #28)
72. #1 SHIFT then #29 S⇔D
73. #2 ALPHA then #29 S⇔D
74. #1 SHIFT then #30 M+
75. #2 ALPHA then #30 M+
76. #1 SHIFT then #31 7
77. #1 SHIFT then #32 8
78. #1 SHIFT then #33 9
79. #1 SHIFT then #34 DEL
80. #1 SHIFT then #35 AC
81. #1 SHIFT then #36 4
82. #1 SHIFT then #37 5
83. #1 SHIFT then #39 ×
84. #1 SHIFT then #40 ÷
85. #1 SHIFT then #41 1
86. #1 SHIFT then #42 2
87. #1 SHIFT then #43 3
88. #1 SHIFT then #44 +
89. #1 SHIFT then #45 -
90. #1 SHIFT then #46 0
91. #1 SHIFT then #47 •
92. #2 ALPHA then #47 •
93. #1 SHIFT then #48 ×10x
94. #2 ALPHA then #48 ×10x
95. #1 SHIFT then #49 Ans

Appendix B — Function finder (A–Z)

Function	Label	Keys
← ENG — engineering notation, shifting the other way	←	#1 → #26
×10^x / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10^x 23	×10x	#48
▶Decimal — show the result as a decimal	≈	#1 → #29
10^x — antilog	10▭	#1 → #17
Absolute value |x|	Abs	#1 → #21
Addition +	+	#44
ALPHA — the next key types its letter/variable (red/green label)	ALPHA	#2
Angle symbol ∠ (polar complex form)	∠	#1 → #19
Ans — recall the last answer into the current expression	Ans	#49
BASE-N — number-base menu (DEC/HEX/BIN/OCT)	BASE	#1 → #43
CALC — evaluate a stored formula for new variable values	CALC	#9
Closing bracket). Closing brackets at the end of a line may be omitted	)	#28
CLR — clear menu (memory / setup / all)	CLR	#1 → #33
CMPLX — complex-number menu	CMPLX	#1 → #42
Colon : (multi-statement separator)	:	#2 → #10
Combinations nCr	nCr	#1 → #40

Comma / argument separator (between function arguments, list items or coordinates)	,	#1 → #28
Common logarithm $\log_{10}$	log	#17
CONST — scientific-constants menu	CONST	#1 → #31
CONV — unit-conversion menu	CONV	#1 → #32
Cosine (current angle unit)	cos	#23
Cube root $\sqrt[3]{}$	$\sqrt[3]{\square}$	#1 → #14
Cube x^3	x^3	#1 → #15
Cursor ▲ — up / previous history line / previous item	▲	#3
Cursor ▶ — move right in the line	▶	#6
Cursor ▼ — down / next history line / next item	▼	#4
Cursor ◀ — move left in the line	◀	#5
Decimal point. Pressing it first gives "0."; a second point in the same number is ignored	•	#47
Definite integral ∫ (numerical, Simpson/adaptive)	$\int\square$	#10
Degrees-minutes-seconds entry/display °′ ″	°′ ″	#20
Degrees-minutes-seconds entry/display °′ ″	←	#1 → #20
DEL — delete the character at the cursor	DEL	#34
Derivative d/dx at a point (numerical)	d/dx	#1 → #10
Digit 0	0	#46
Digit 1	1	#41
Digit 2	2	#42
Digit 3	3	#43
Digit 4	4	#36
Digit 5	5	#37

Digit 6	**6**	**#38**
Digit 7	**7**	**#31**
Digit 8	**8**	**#32**
Digit 9	**9**	**#33**
Division ÷	**÷**	**#40**
DRG — choose/convert the angle unit (Deg / Rad / Grad)	**DRG▶**	**#1 → #49**
ENG — engineering notation (exponent multiple of 3)	**ENG**	**#26**
Equals — complete the calculation	**=**	**#2 → #9**
Euler's number e (2.71828…)	**e**	**#2 → #48**
e^x — exponential	**e▭**	**#1 → #18**
Execute / = — evaluate the line and show the result (EXE / ENTER / =)	**=**	**#50**
Factorial n! (also on non-integers via the gamma function where supported)	**x!**	**#1 → #11**
Fraction template ▭/▭ (numerator over denominator)	**▭/▭**	**#13**
Hyperbolic prefix — the next sin/cos/tan becomes sinh/cosh/tanh	**hyp**	**#21**
Imaginary unit i (complex numbers)	**i**	**#2 → #26**
INS — insert mode at the cursor	**INS**	**#1 → #34**
Inverse cosine $\cos^{-1}$	**$\cos^{-1}$**	**#1 → #23**
Inverse sine $\sin^{-1}$	**$\sin^{-1}$**	**#1 → #22**
Inverse tangent $\tan^{-1}$	**$\tan^{-1}$**	**#1 → #24**
log with any base: $\log_a(b)$	**log▭▭**	**#12**
M+ — add the value to independent memory M	**M+**	**#30**

M− — subtract the value from memory M	**M−**	**#1 → #30**
MATRIX menu	**MATRIX**	**#1 → #36**
Mixed-number template (whole, numerator, denominator)	**▭⌟▭/▭**	**#1 → #13**
MODE — choose calculation mode	**MENU**	**#7**
Multiplication ×	**×**	**#39**
n-th root $\sqrt[x]{y}$	**▭√▭**	**#1 → #16**
Natural logarithm ln	**ln**	**#18**
Negative sign (−) — makes the next number negative. Different from the subtraction key	**(−)**	**#19**
OFF — power off (state is lost; press ON to restart)	**OFF**	**#1 → #35**
ON — power on	**ON**	**#8**
ON / AC — power on, or clear all	**AC**	**#35**
Opening bracket (	**(**	**#27**
Percent %	**%**	**#1 → #27**
Permutations nPr	**nPr**	**#1 → #39**
Polar → rectangular Rec(r,θ)	**Rec**	**#1 → #45**
Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)	**x▭**	**#16**
Random integer in a range	**RanInt**	**#2 → #47**
Random number 0–1	**Ran#**	**#1 → #47**
RCL — recall a memory / variable	**RCL**	**#25**
Reciprocal x^{-1} (1/x)	**x^{-1}**	**#11**

Rectangular → polar Pol(x,y)	Pol	#1 → #44
Round to the displayed precision	Rnd	#1 → #46
S⇔D — toggle the result between exact (fraction, √, π) and decimal	S⇔D	#29
SETUP — angle unit, number format, display options	SETUP	#1 → #7
SHIFT / 2nd — the next key uses its upper (shift-coloured) function	SHIFT	#1
Sine (uses the current angle unit)	sin	#22
SOLVE — numerically solve an equation for a variable	SOLVE	#1 → #9
Square root √	√▭	#14
Square x^2	x^2	#15
STAT — statistics menu	STAT	#1 → #41
STO — store the value into a memory / variable (press the target next)	STO	#1 → #25
Subtraction −	−	#45
Summation Σ over an integer range	Σ	#1 → #12
Tangent (current angle unit)	tan	#24
Variable A (store with STO/→, recall by name)	A	#2 → #19
Variable B (store with STO/→, recall by name)	B	#2 → #20
Variable C (store with STO/→, recall by name)	C	#2 → #21
Variable D (store with STO/→, recall by name)	D	#2 → #22
Variable E (store with STO/→, recall by name)	E	#2 → #23
Variable F (store with STO/→, recall by name)	F	#2 → #24

Variable M (store with STO/ →, recall by name)	**M**	**#2 → #30**
Variable X (store with STO/ →, recall by name)	**X**	**#2 → #28**
Variable Y (store with STO/ →, recall by name)	**Y**	**#2 → #29**
VECTOR menu	**VECTOR**	**#1 → #37**
π (3.14159265…) as an exact constant	**π**	**#1 → #48**

Appendix C — Glossary

Ans
The result of the last calculation, reusable in the next.

AOS
Algebraic Operating System — operator precedence on a one-line display.

RPN
Reverse Polish Notation — operands first, then the operation; no = key.

Stack
The RPN registers X, Y, Z, T where numbers wait for operations.

TVM
Time Value of Money — n, i, PV, PMT, FV.

Natural display
Shows fractions, roots and powers as printed in textbooks.

Shift / 2nd
Prefix key that selects the function printed above a key.

Alpha
Prefix key that types the letter printed beside a key.

Radian
Angle unit; 2π rad = 360°.

Grad
Angle unit; 400 grad = 360°.

Engineering notation
Scientific notation with exponents that are multiples of 3.

Mantissa
The digits of a number in scientific notation.

Memory (M)

An independent register for running totals.

Mark-up

Increase over cost; margin is profit as a percent of selling price.

Grand total (GT)

Sum of all totals since the last clear.

nCr / nPr

Combinations / permutations of n items taken r at a time.

Factorial

n! = 1×2×…×n.

Reciprocal

1/x.

Standard deviation

Spread of data around the mean (σ population, s sample).

Key number

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

Appendix D — Model notes & limits

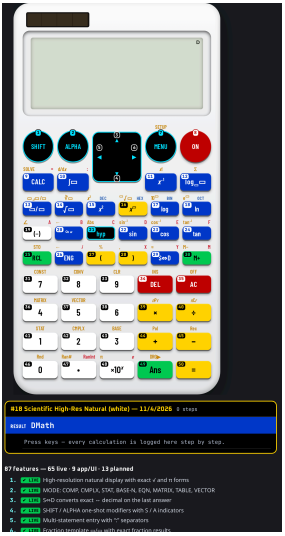
High-resolution 4-line natural display. MENU opens the calculation-mode menu (Calculate, Complex, Base-N, Matrix, Vector, Statistics, Table, Equation). Spreadsheet, Inequality, Ratio and Distribution modes are not included.

Planned (not yet working)

- Spreadsheet app
- Verify (equation true/false)
- Ratio solver
- Inequality solver
- QR-code result export
- Prime factorisation
- Recurring decimals
- Inverse-normal distribution
- Exam mode (locks memory & shows indicator)
- Answer history export to text
- Speech read-out of results
- Colour-blind safe key theme
- Custom shortcut keys

U-45 CALCULATORS models are independent software recreations; they are not affiliated with any calculator manufacturer and carry no third-party trademarks.

Appendix E — Quick-reference card



#1 SHIFT	**#2** ALPHA	**#3** ▲
#4 ▼	**#5** ◀	**#6** ▶
#7 MENU ⇧ SETUP	**#8** ON	**#9** CALC ⇧ SOLVE
#10 ∫▭ ⇧ d/dx	**#11** x^{-1} ⇧ x!	**#12** log▭▭ ⇧ Σ
#13 ▭/▭ ⇧ ▭⌟▭/▭	**#14** √▭ ⇧ $\sqrt[3]{\square}$	**#15** x^2 ⇧ x^3
#16 x▭ ⇧ ▭√▭	**#17** log ⇧ 10▭	**#18** ln ⇧ e▭
#19 (–) ⇧ ∠	**#20** °′″ ⇧ ←	**#21** hyp ⇧ Abs
#22 sin ⇧ $\sin^{-1}$	**#23** cos ⇧ $\cos^{-1}$	**#24** tan ⇧ $\tan^{-1}$
#25 RCL ⇧ STO	**#26** ENG ⇧ ←	**#27** (⇧ %
#28) ⇧ ,	**#29** S⇔D ⇧ ≈	**#30** M+ ⇧ M–
#31 7 ⇧ CONST	**#32** 8 ⇧ CONV	**#33** 9 ⇧ CLR
#34 DEL ⇧ INS	**#35** AC ⇧ OFF	**#36** 4 ⇧ MATRIX
#37 5 ⇧ VECTOR	**#38** 6	**#39** × ⇧ nPr
#40 ÷ ⇧ nCr	**#41** 1 ⇧ STAT	**#42** 2 ⇧ CMPLX
#43 3 ⇧ BASE	**#44** + ⇧ Pol	**#45** – ⇧ Rec
#46 0 ⇧ Rnd	**#47** • ⇧ Ran#	**#48** ×10x ⇧ π
#49 Ans ⇧ DRG▶	**#50** =	

U-45 CALCULATORS · Calculator #18 · cut out and keep